

Diversity in Living Organisms

1. Match column-I with column-II and select the correct option :-

Column-I	Column-II
(a) Family	(i) tuberosum
(b) Kingdom	(ii) Polymoniales
(c) Order	(iii) Solanum
(d) Specific epithet	(iv) Plantae
(e) Genus	(v) Solanaceae

(1) (a)–v, (b)–iv, (c)–ii, (d)–i, (e)–iii

(2) (a)–v, (b)–iv, (c)–i, (d)–iii, (e)–ii

(3) (a)–i, (b)–ii, (c)–iii, (d)–v, (e)–iv

(4) (a)–ii, (b)–i, (c)–iii, (d)–iv, (e)–v

2. Which of the following statements are correct?

(A) Genera are aggregates of closely related species.

(B) Families are characterised on the basis of vegetative characters only.

(C) Order and other higher categories are identified on the basis of aggregates of characters.

(1) A and B

(3) A, B, C

(2) B and C

(4) A and C

3. Read the following statements carefully about binomial nomenclature and identify the incorrect statement :-

(1) Biological names are generally in Latin and printed in italics.

(2) The first component in a biological name represents the species while second component denotes the genus.

(3) Both the words in a biological name, when handwritten, are separately underlined or printed in italics to indicate their Latin origin

(4) The first word denoting genus starts with a capital letter while specific epithet starts with a small letter

4. Find the correct statements from the followings:-

(A) Each rank or taxon, in fact represents a unit of classification

(B) Every step in taxonomic hierarchy represents a rank or category

(C) A taxonomic category is a part of overall taxonomic hierarchy

(D) Taxonomic categories are not distinct biological entities, they are only morphological aggregates

Options :-

(1) A, B & D are correct

(2) A, C & D are correct

(3) A, B & C are correct

(4) Only A and B are correct

5. Given below are two statements labelled as

Assertion (A) and Reason (R)

Assertion : Characterisation, identification, classification & nomenclature are the processes that are basic to taxonomy.

Reason : Systematics takes into account evolutionary relationships between organisms.

(1) Both Assertion & Reason are correct and the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are correct but Reason is not a correct explanation of the Assertion.

(3) Assertion is correct but the Reason is incorrect.

(4) Both Assertion & Reason are incorrect.

6. Statement-1 : Yeast is a multicellular fungus.

Statement-2 : Penicillium is an unicellular fungus.

Statement-3 : Albugo is a parasitic fungus on mustard.

- (1) Only statement-1 and statement-2 are correct
- (2) All the above statements are incorrect
- (3) Only statement-3 is correct
- (4) Both statement-1 and statement-3 are correct

7. Given below are two statements labelled as Assertion (A) and Reason (R)

Assertion : Euglena can be placed in the plant kingdom due to the presence of chlorophyll.

Reason : Euglena cannot be classified on the basis of two kingdom system of classification.

- (1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
- (2) If both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.
- (3) If Assertion is true but Reason is false.
- (4) If both Assertion and Reason are false.

8. Choose the wrong statement for chemosynthetic autotroph bacteria

- (1) They oxidise various inorganic substrate such as nitrates, nitrites & ammonia and use the released energy for their ATP production
- (2) They play great role in recycling nutrient like nitrogen phosphorous, iron & sulphur
- (3) For their energy production they utilize solar energy
- (4) They can prepare their food from inorganic substrate.

9. Select the correct statement.

- (1) Viroids have double stranded RNA
- (2) RNA of viroids have high molecular weight than viruses
- (3) Mumps and Herpes are viral diseases
- (4) The name virus was given by D.J. Ivanowsky

10. Members of phycomycetes are found in

- i. Aquatic habitats
- ii. On decaying wood
- iii. Moist and damp places
- iv. As obligate parasites on plants

- (1) None of the above
- (2) i and iv
- (3) ii and iii
- (4) All of the above

11. Given below are two statements:

Statement-I : Fungi and members of the Monera and Protista having cell walls have now been excluded from Plantae, although according to earlier classifications they were placed in the same kingdom.

Statement-II : The earliest systems of classification used only gross superficial morphological characters such as habit, colour, number and shape of leaves etc.

- (1) Both Statement-I and Statement-II are correct.
- (2) Both Statement-I and statement-II are incorrect.
- (3) Statement-I is correct but Statement-II is incorrect.
- (4) Statement-I is incorrect but Statement II is correct.

12. Read Which of the following spores are produced endogenously?

- (1) Zoospores and Conidia
- (2) Conidia and aplanospores
- (3) Aplanospores and zoospores
- (4) Aplanospore, zoospores and conidia

13. When there are two haploid nuclei per cell in some fungi before the formation of diploid, this stage is called

- (1) Diplotene
- (2) Diplophase
- (3) Dikaryophase
- (4) Dikaryote

14. With respect to fungal sexual cycle, choose the correct sequence of events

- (1) Karyogamy, Plasmogamy and Meiosis
- (2) Meiosis, Plasmogamy and Karyogamy
- (3) Plasmogamy, Karyogamy and Meiosis
- (4) Meiosis, Karyogamy and Plasmogamy

15. How many of the following are correct about plantae-

I. Cells have eukaryotic structure

II. Prominent chloroplast

III. Cellulosic cell wall

IV. Life cycle has three distinct phases

V. Show alternation of generation

- (1) One
- (2) Two
- (3) Three
- (4) Four

16. Match column I with column II and choose the right option

Column I

Column II

- | | |
|--------------------------|-------------------------|
| A. Ascomycetes | (i) Rhizopus |
| B. Basidiomycetes | (ii) Penicillium |
| C. Deuteromycetes | (iii) Ustilago |
| D. Phycomycetes | (iv) Alternaria |

- (1) A-(iv), B-(iii), C-(i), D-(ii)
- (2) A-(ii), B-(iii), C-(iv), D-(i)
- (3) A-(iv), B-(i), C-(ii), D-(iii)
- (4) A-(iii), B-(iv), C-(ii), D-(i)

17. Given below are two statements:

Statement-1 : Linnaeus classified plants into trees, shrubs and herbs, on the basis of morphological characters.

Statement-2 : Aristotle divided animals into groups - Anaima and Enaima.

- (1) Only statement-1 is correct
- (2) Only statement-2 is correct
- (3) Both statement-1 and statement-2 are correct
- (4) Both statement-1 and statement-2 are incorrect

18. Read Which of the following statement(s) is/are correct about ascomycetes ?

(i) Neurospora, which is used in biochemical and genetic work is a member of this class.

(ii) They are mostly multicellular, e.g., Yeast, or rarely unicellular, e.g., Penicillium.

(iii) They are saprophytic, decomposers, parasitic or coprophilous.

(iv) Some examples are Aspergillus, Claviceps and Neurospora.

- (1) Both (i) and (ii)
- (2) Only (ii)
- (3) (i), (iii) and (iv)
- (4) All of these

19. Select the incorrect statement.

- (1) Nostoc and Anabaena have heterocysts for nitrogen fixation
- (2) Cyanobacteria often form blooms in polluted water bodies
- (3) Heterotrophic bacteria are more abundant in nature
- (4) The cell wall of Mycoplasma is made up of chitin

20. The given statements are some characters of a particular group of Kingdom Protista.

(i) Most of them are fresh water organisms found in standing water.

(ii) They have a protein rich layer (called pellicle) which makes their body flexible.

(iii) They have two flagella, a short and a long one.

(iv) Though they are photosynthetic in the presence of sunlight, but in the absence of sunlight they behave like heterotrophs by preying on other smaller organisms.

Identify the correct group on the basis of these characters.

- (1) Protozoans
- (2) Chrysophytes
- (3) Slime moulds
- (4) Euglenoids

21. Identify the correct set of statements :

- (a) One of the gymnosperms, the gaint red wood tree sequoia is one of the tallest tree species.**
- (b) Gymnosperms are heterosporous cryptogams.**
- (c) Gymnosperms are evolutionarily last archegoniate plant group.**
- (d) Pteridophytes are evolutionarily last antheridial plant group.**
- (e) The stems are unbranched in Cycas and branched in Pinus and Cedrus.**

- (1) (a), (b) and (d) only
- (2) (c), (d) and (e) only
- (3) (a), (b), (c) and (d) only
- (4) (a), (c), (d) and (e) only.

22. Which of the following statements is wrong w.r.t. bryophytes?

- (1) They are found in damp, shady and humid localities
- (2) They are called amphibians of plant kingdom
- (3) Gametophyte is diploid and dominant plant body
- (4) Bryophytes are homosporous

23. Protonema stage is –

- (1) Creeping, green unbranched and frequently filamentous stage
- (2) Prostate, green, branched and frequently filamentous stage
- (3) Creeping, green, branched and frequently filamentous stage
- (4) Prostate, non – green, unbranched and frequently stage

24. Identify the correct set of statements :

- (a) In spirogyra sporophytic generation is represented only by one called zygote.**
- (b) Fucus is a brown alga shows diplontic life cycle.**
- (c) Bryophytes and pteridophytes show diplontic life cycle.**
- (d) Ectocarpus, polysiphonia, kelps are haplodiaplontic.**

(e) Gymnosperms and angiosperms show haplontic life cycle.

- (1) (a), (b) and (e) only
- (2) (a), (b), (d) and (e) only
- (3) (a), (b) and (d) only
- (4) (b) and (d) only

25. Hydrocolloids of red algae

- (a) Are water holding substances**
- (b) Also includes agar**
- (c) Are produced by Gelidium and Gracilaria**

Which of the above statements are true?

- (1) All (a), (b) and (c)
- (2) (a) and (b) only
- (3) (b) and (c) only
- (4) (a) and (c) only

26. Find the odd one out w.r.t. chlorophyceae

- (1) Storage bodies are called pyrenoids
- (2) Pyrenoid are composed of protein and starch
- (3) Ectocarpus, Dictyota and Fucus are members of this class
- (4) Chlorophyll a, b and carotene are pigments found in chlorophyceae

27. Consider the following characters :-

- (A) Formation of only one functional megaspore in a megasporangium**
- (B) Formation of hard covering around megasporangium**
- (C) Development of embryo from zygote within the female gametophyte**
- (D) Retention of megaspore inside the megasporangium**

These were very essential events occurred during the course of evolution for the phenomenon of:-

- (1) Heterospory
- (2) Seed habit
- (3) Fruit formation
- (4) Covered seed formation

28. Given below are two statements:

Statement-I : Most of the members of chlorophyceae have one or more storage bodies called pyrenoids located in the chloroplasts.

Statement-II : Cell wall of members of rhodophyceae is made up of cellulose, pectin and polysulphate esters.

Choose the most appropriate answer from the options given below:

- (1) Both Statement-I and Statement-II are correct.
- (2) Both Statement-I and statement-II are incorrect.
- (3) Statement-I is correct but Statement-II is incorrect.
- (4) Statement-I is incorrect but Statement II is correct.

29. The Which of the given is a feature of 'Phaeophyceae'?

- (1) Cell wall is composed of cellulose, pectin and polysulphated esters
- (2) Floridian starch is stored food
- (3) They have carrageen and agar in cell wall
- (4) Produce flagellated spores and gametes

30. Given below are two statements:

Statement-I : Selaginella and Salvinia are homosporous pteridophytes.

Statement-II : Gymnosperms are tracheophytes, embryophytes, heterosporous, spermatophytes phanerogams.

Choose the most appropriate answer from the options given below:

- (1) Both Statement-I and Statement-II are correct.
- (2) Both Statement-I and statement-II are incorrect.
- (3) Statement-I is correct but Statement-II is incorrect.
- (4) Statement-I is incorrect but Statement II is correct

31. Classification system given by Linnaeus and Hutchinson respectively are

- (1) Natural and artificial
- (2) Artificial and natural
- (3) Artificial and phylogenetic
- (4) Phylogenetic and artificial

32. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : In heterosporous pteridophytes, gametophyte is dioecious.

Reason (R) : The female gametophytes in heterosporous pteridophytes, are retained on parent sporophyte for variable periods.

Choose the most appropriate answer from the options given below :

- (1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
- (2) (A) is correct but (R) is not correct.
- (3) (A) is not correct but (R) is correct.
- (4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

33. Given below are two statements:

Statement-I : The cones bearing megasporophyll with ovules are female cone

Statement-II : The strobili bearing microsporangia are called male cone

Choose the most appropriate answer from the options given below :

- (1) Both stated statement are correct
- (2) Both stated statement are incorrect
- (3) Statement-I is correct while statement II is incorrect
- (4) Statement-I is incorrect while statement-II is correct

34. Given below are two statements:

Assertion (A) : Artificial system of classification separated closely related species.

Reason (R) : Artificial system of classification was based on a few characteristics.

Choose the most appropriate answer from the options given below :

- (1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
- (2) (A) is correct but (R) is not correct.
- (3) (A) is not correct but (R) is correct.
- (4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

35. Mosses

- (1) Have less developed sporophyte than liver worts
- (2) Exhibit creeping, green and branched filamentous stage called protonema, which is their juvenile phase
- (3) Show diplontic life cycle
- (4) May asexually reproduce by gammae

36. Aedes, Ancylostoma and Aplysia are similar to each other as they all have a

- (1) Body cavity that is completely lined by mesoderm
- (2) Well-developed circulatory system
- (3) Ventral heart
- (4) Similar level of body organisation

37. In which of the following organisms, any plane passing through the central axis of the body, divides the organism into two identical halves?

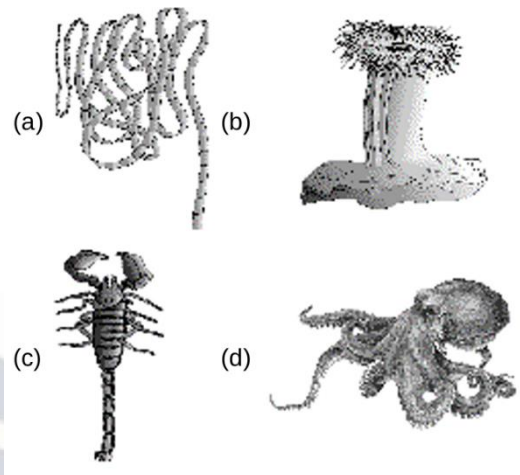
- a. Larvae of Sepia
- b. Adult form of Cucumaria
- c. Adult form of Ctenoplane
- d. Adult form of Meandrina
- e. Larvae of Echinus

Select the correct option.

- (1) a, b and c
- (2) b, c and d
- (3) c, d and e
- (4) b, d and e

38. The figures given below represent four animals.

Select the correct answer with respect to a common characteristic of two of these animals.



- (1) (a) and (c) are metamerically segmented
- (2) (b) and (d) have cnidoblasts for self-defence
- (3) (c) and (d) possess a true coelom
- (4) (a) and (b) are radially symmetrical

39. Given below is a list of features:

- (i) Ear is represented by tympanum.
- (ii) Oviparity is seen.
- (iii) Eyes have eyelids.
- (iv) Skin is moist without scales.

The above given features are associated with members of class

- (1) Osteichthyes
- (2) Chondrichthyes
- (3) Amphibia
- (4) Reptilia

40. Consider the following statements.

Statement A: Hag-fish is a jawless true fish that emigrate to rivers for spawning.

Statement B: In annelids, the neural system consists of paired ganglia connected by lateral nerves to a single ventral nerve cord.

Choose the correct option.

- (1) Both statements A and B are incorrect
- (2) Both statements A and B are correct
- (3) Only statement A is correct
- (4) Only statement B is correct

41. Choose the incorrect option.

- (1) Complete digestive system - Two openings, mouth and anus
- (2) Incomplete digestive system - Single opening system
- (3) Open circulatory system - Blood is circulated through tubes
- (4) Closed circulatory system - Arteries veins and capillaries present

42. Which of the following pairs are correctly matched?

- (A) Poikilothermous - Birds and mammals
- (B) Agnatha - Petromyzon and Myxine
- (C) Tunicates - Ascidia, Salpa and Doliolum
- (D) Cartilaginous fishes - Betta and Labeo

- (1) A and B
- (2) A, B and C
- (3) B and C
- (4) A, B and D

43. Choose the incorrect option

- (1) True coelom is a body cavity which arises as a cavity in the embryonic mesoderm.
- (2) Digestive cavity is found in acoelomates, pseudocoelomates as well as coelomates.
- (3) The body cavity of arthropods and non-cephalopod molluscs is called haemocoel.
- (4) There is no cavity between the body wall and gut wall in echinoderms.

44. Given below are two statements:

Statement-I : In cephalochordate notochord is persistent throughout their life.

Statement-II : Branchiostoma is an example of cephalochordata

Choose the most appropriate answer

- (1) Both Statement-I and II are incorrect.
- (2) Statement-I is correct but statement-II is incorrect.
- (3) Statement-I is incorrect & Statement II is correct.
- (4) Both Statement-I and Statement-II are correct.

45. Which of the following animals are true coelomates with bilateral symmetry?

- (1) Adult echinoderms
- (2) Aschelminthes
- (3) Platyhelminthes
- (4) Annelids

46. A new animal was discovered and brought to the lab for scientific observation and study. The following characteristic features were documented about it.

- (i) The adults were tetrapods
- (ii) A lot of difference was observed between larvae and adults.
- (iii) The tails were absent in the adults, which the larvae had.
- (iv) Adults could live both on land and water.

The organism most likely belongs to

- (1) Arthropoda
- (2) Pisces
- (3) Protochordata
- (4) Amphibia

47. How many of the features listed below are correct w.r.t. phylum porifera?

- (i) Internal cavities are lined with food filtering flagellated cells
- (ii) Body is made up of meshwork of cells
- (iii) Mostly asymmetrical
- (iv) Water vascular system

- (1) None
- (2) One
- (3) Two
- (4) Three

48. Select the correct match w.r.t. the taxonomic position of lamprey.

- (1) Phylum : Vertebrata
- (2) Sub-phylum : Gnathostomata
- (3) Division : Agnatha
- (4) Super class : Cyclostomata

49. Choose the organism for which all of the below listed characteristics are true.

- (i) Presence of air bladder
- (ii) Curved tail
- (iii) Presence of long snout
- (iv) Bony endoskeleton

Select the correct option.

- (1) Hippocampus
- (2) Catla
- (3) Scoliodon
- (4) Pristis

50. Given below are two statements:

Assertion (A): Struthio and Salamandra can regulate their body temperature as per the ambient environment.

Reason (R): The members belonging to the class Amphibia and Aves are homeotherms.

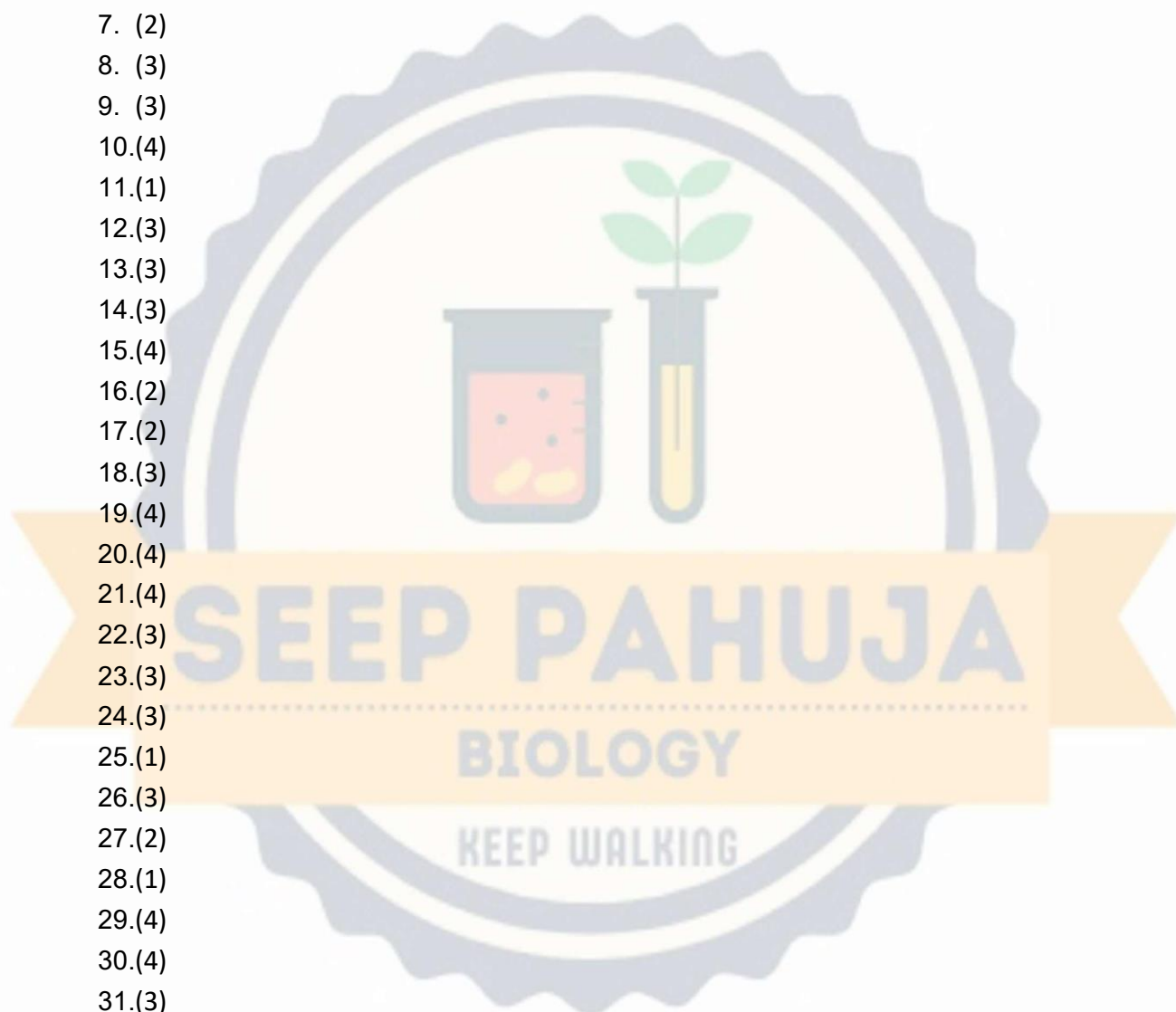
In the light of above statements, choose the correct option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true; (R) is false
- (4) Both (A) and (R) are false



Answer Key

1. (1)
2. (4)
3. (2)
4. (3)
5. (2)
6. (3)
7. (2)
8. (3)
9. (3)
- 10.(4)
- 11.(1)
- 12.(3)
- 13.(3)
- 14.(3)
- 15.(4)
- 16.(2)
- 17.(2)
- 18.(3)
- 19.(4)
- 20.(4)
- 21.(4)
- 22.(3)
- 23.(3)
- 24.(3)
- 25.(1)
- 26.(3)
- 27.(2)
- 28.(1)
- 29.(4)
- 30.(4)
- 31.(3)
- 32.(1)
- 33.(1)
- 34.(4)
- 35.(2)
- 36.(4)
- 37.(2)
- 38.(3)
- 39.(3)
- 40.(1)



- 41.(3)
- 42.(3)
- 43.(4)
- 44.(4)
- 45.(4)
- 46.(4)
- 47.(4)
- 48.(3)
- 49.(1)
- 50.(4)

